Poornachandra G.

Bioinformatics & AI Engineer | Precision Medicine & Agentic AI Systems

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PROFESSIONAL SUMMARY

Bioinformatics & AI Engineer with 10+ years of technical experience designing, building, and deploying end-to-end biological foundation-model systems, agentic workflows, and reproducible multi-omics pipelines. Expertise combining LLMs with validated bioinformatics tools, retrieval-augmented generation (RAG), and systematic evaluation frameworks (Promptfoo, DeepEval, Ragas) to minimize hallucination risk and assess biological relevance. Proven track record in high-throughput NGS pipeline engineering (Nextflow, AWS HealthOmics, Docker, GPU acceleration), clinical variant interpretation (ACMG/AMP), and large-scale public data curation (TCGA, ClinVar, gnomAD) for precision oncology and translational research.

SELECTED HIGHLIGHTS & IMPACT

- Clinical NGS Pipelines: Built and productionized end-to-end NGS pipelines (WGS/WES/TES, RNA-Seq, CNV, SV) used in clinical and research workflows; significantly reduced sample turnaround time through automated QC and reporting.
- Variant Interpretation Workflows: Architected variant annotation and prioritization workflows (ACMG/AMP/ASCO/CAP) integrating multiple public and proprietary annotations to enable actionable clinical decision support.
- Biomarker Discovery & ML Initiatives: Led ML initiatives for biomarker discovery and patient stratification: feature engineering from multi-omics inputs, model validation, and deployment in production environments.
- Phenotype-Based Rare Disease Prioritization: Architected and built a tertiary analysis pipeline for phenotype-based gene prioritization, variant selection, and reporting in rare diseases. Presented this work and secured the top rank at the Bangalore Tech Summit (2025) as an independent researcher.
- Bioinformatics Team Mentorship & Hiring: Mentored and grew a bioinformatics team; led technical hiring, performance reviews, and cross-functional training.
- AI/LLM Engineering & Observability: Architected independent AI projects spanning LLM tooling, multi-agent frameworks, clinical AI apps, and developer infrastructure. Established AI evaluation and observability frameworks (Promptfoo, DeepEval, Ragas) for accuracy benchmarking, hallucination analysis, and retrieval quality validation.
- Internal Web Applications & Dashboards: Delivered internal web applications and dashboards (Streamlit, Gradio, FastAPI) to deploy AI-based applications for cross-functional stakeholders.

SKILLS & TECHNOLOGIES

Pipeline Engineering	Nextflow • Benchmarking • Reproducibility • Seurat • GATK • Sentieon • DRAGEN • NVIDIA Parabricks • FreeBayes • DeepVariant • DeepSomatic • PureCN • CNVKit • VEP • ANNOVAR • QC Automation • Kallisto • Salmon • DESeq2 • edgeR • Scanpy • GSEA
Programming & Engineering	Python • R • Git • CI/CD • Bash • FastAPI • SQLite • PostgreSQL • REST APIs • Uvicorn
Cloud, Platforms & Infrastructure	Docker • HPC • AWS (EC2, S3, Batch, HealthOmics) • Azure AI Foundry • DNAnexus • BaseSpace • Galaxy • GenePattern • Terra/FireCloud • Illumina Connected Analytics • Conda/Mamba • Jupyter • MLOps
Machine Learning & AI	R • Supervised/Unsupervised Learning • Feature Engineering • LightGBM • XGBoost • PyTorch • TensorFlow • scikit-learn • PyCaret • AutoML • PySpark • SHAP • scipy • pandas • NumPy
LLMs & Agentic Systems	Fine-Tuning • LangChain • LangGraph • CrewAI • Multi-Agent Systems • RAG Pipelines • GraphRAG • Prompt Engineering • Context Engineering • Quantization • LoRA/QLoRA • AI Evaluation & Observability (Promptfoo, DeepEval, Ragas, LangSmith, Langfuse) • AI Validation (LLM-as-a-Judge, Red Teaming, Hallucination Analysis) • LiteLLM • MCP (Model Context Protocol) • OpenAI / Anthropic / Gemini APIs • Ollama • W&B
Applied Bioinformatics & Clinical-omics	WGS • WES • TES • RNA-Seq • scRNA-Seq • Spatial Transcriptomics • Liquid Biopsy • Pharmacogenomics • Companion Diagnostics • Variant Interpretation (ACMG/AMP/ASCO/CAP) • MSI • HRD • TMB • CNV • SV • Rare Disease Genomics • Phenotype-based Gene Prioritization
NLP, OCR & Document AI	Hugging Face Transformers • LayoutLMv3 • BioBERT • scispaCy • NLTK • Tesseract • EasyOCR • PyMuPDF • Docling • unstructured.io • OpenCV
Visualization & UI	Streamlit • Gradio • Plotly • Matplotlib • Seaborn • Next.js • HTML/CSS/JS
Genai Developer Tools	Claude Code • GitHub Copilot • Cursor • Windsurf • Continue • Cline • Aider • Goose
Leadership & Delivery	Team Building • Technical Hiring • Mentoring • Technical Roadmapping • Stakeholder Communication • HIPAA Compliance • Agile / Scrum • Risk Management • JIRA • Cross-Functional Training

PROFESSIONAL EXPERIENCE

Cellworks - Lead Scientist Bioinformatics
2024 - Present

- Led the Bioinformatics R&D team: defined technical roadmap, recruited and mentored scientists, drove project prioritization, and aligned strategy with executive leadership.
- Designed multi-agent orchestration workflows for vendor order management, coordinating specialized agents, external tools, and contextual memory with human-in-the-loop validation at critical decision points.
- Designed and maintained golden datasets, benchmark queries, and expected-answer sets that powered repeatable evaluation of retrieval quality, response accuracy, hallucination reduction, and regression testing across GenAI applications.
- Built reusable prompt libraries and test suites (zero-shot, few-shot, role-based, multi-turn, edge-case) using Promptfoo and DeepEval to validate GenAI feature accuracy and consistency before release.
- Built multi-stage annotation and prioritization pipelines applying AMP/ASCO/CAP guidelines across public and proprietary annotation sources.
- Designed and implemented input ingestion, monitoring, and reporting frameworks that reduced manual review time by 20%.
- Established risk management practices for bioinformatics pipelines - identified potential failure points, data-quality risks, and downstream repercussions, and implemented mitigation strategies.
- Designed structured upskilling sessions leveraging supplemental projects beyond core deliverables, improving engagement, critical thinking, and problem-solving across team members.

Cellworks - Senior Scientist, Reporting Manager
2021 - 2024

- Developed comprehensive workflows for microsatellite instability (MSI), homologous recombination deficiency (HRD), mutational signatures, tumor mutational burden (TMB), and indel burden.
- Built high-throughput variant calling stacks on AWS following GATK best practices, including mutation and CNV calling, and biomarker extraction.
- Built and led the bioinformatics team: recruited, trained, mentored, and reviewed the work of 4 members; created custom training protocols that accelerated onboarding and competency development.
- Designed input-agnostic workflows supporting multiple formats (FASTQ, BAM, VCF, structured reports) across diverse commercial vendors (4BC, Tempus, Caris, Foundation Medicine, Alacris).
- Built cancer driver and copy number variation (CNV) vs. mRNA expression correlation databases for 30+ cancer types using public datasets and computational tools.
- Developed automated multi-omics interpretation pipelines for high-impact clinical trial datasets from biopharma partners including Takeda and Genentech.
- Created an in-house Variant of Uncertain Significance (VUS) reclassification pipeline using an Ensembl-based approach, improving classification accuracy by >40%.
- Analyzed single-cell RNA-seq data from Human Protein Atlas normal-tissue samples with Seurat, Scanpy, and limma to establish in-silico control baselines for distinguishing cancer-specific expression signatures.
- Applied CIBERSORTx and MuSiC bulk RNA-seq deconvolution to resolve cancer-specific markers within heterogeneous tumor samples and improve multi-omics interpretation.

Cellworks - Senior Scientist, Bioinformatics
2018 - 2021

- Developed and benchmarked secondary analysis pipelines (Illumina DRAGEN, NVIDIA Parabricks, Sentieon), optimized for turnaround time, resource cost, and variant calling accuracy.
- Led custom panel design for liquid-biopsy samples; collaborated with Cleveland Clinic Florida to build a machine learning pipeline using longitudinal genomic data to predict early relapse in Melanoma.
- Built an analytical framework to process pediatric cancer cases; served as primary domain owner for pediatric hematological malignancies.
- Designed and implemented a Panel of Normals (PoN) workflow for CNV and mutation calling using ~100 cases, improving support for tumor-only clinical analyses.
- Created multiple tertiary analysis workflows (GSEA, pathway enrichment, PCA, statistical models, ML pipelines) for biomarker discovery and treatment-response characterization.
- Developed multi-omics interpretation pipelines for the Foundation Medicine clinical genomic dataset (18,000+ patient cases).
- Compiled a comprehensive diagnostic assay and genomic test inventory across 88 cancer types based on NCCN, FDA, and international clinical guidelines.

Cellworks - Scientist, Clinical Translation
2017 - 2018

- Curated therapeutic rationale libraries to support computational biology predictions, reducing turnaround time for live clinical case analyses by >40%.
- Analyzed 200+ complex cancer cases, gaining deep expertise in target selection using biological network models and drug interaction mechanisms in precision oncology.
- Conducted rigorous benchmarking of external sequencing pipelines (DNAnexus, BeOne, MedGenome) to establish analytical validity, confidence, and cross-platform reproducibility.
- Developed an in-house variant annotation and filtering pipeline with a custom tier classification system, lowering processing costs and enabling clinical interpretability.

Cellworks - Associate Scientist 2015 - 2017

- Processed, interpreted, and harmonized multi-omics outputs (karyotyping, aCGH, IHC, somatic mutations) for >500 cases, translating insights into automated data-harmonization ETL pipelines.
- Curated and automated retrospective in-house dataset creation by integrating patient and cell-line data from >10,000 GDSC, TCGA, and published clinical studies.
- Standardized genomic data interpretation from external medical partners including Columbia University, University of Florida, Cedars-Sinai, Mayo Clinic, and Washington University.
- Automated report-generation workflows using reproducible pipelines and templates, reducing preparation time from ~15 hours to just minutes per week.

KEY PROJECTS & INNOVATIONS

Phenotype-Driven Rare Disease Prioritization Pipeline Nextflow • Python • HPO (Human Phenotype Ontology) • ACMG Guidelines • Variant Prioritization

- End-to-end tertiary analysis pipeline for phenotype-based gene prioritization, variant selection, and clinical reporting in rare genetic disorders. Awarded Top Rank at Bangalore Tech Summit 2025.

PathoParse NLP • BioBERT • Document AI • LayoutLMv3 • Tesseract OCR

- LLM-assisted and rule-based pipelines to extract structured clinical metadata and biomarker stages from unstructured pathology and histology reports for downstream analysis.

VarEffex Multi-Agent • Literature Mining • PubMed API • LangGraph • ACMG

- Multi-agent framework that autonomously searches biomedical literature to extract variant-function associations, functional impacts, and biomarker evidence under ACMG/AMP criteria.

EDUCATION

B.Tech in Biotechnology — JNTUA College of Engineering Pulivendula (JNTUA CEP) (*2015*)

HONORS & AWARDS

- **Bangalore Tech Summit 2025 Top Rank (Independent Research) (2025)**: Awarded Top Rank at Bangalore Tech Summit 2025 as an independent researcher for architecting a phenotype-based gene prioritization and tertiary analysis pipeline for rare disease diagnostics.

SELECTED PUBLICATIONS

- Castro M, ..., Poornachandra G, et al. Combination chemotherapy versus temozolomide for patients with methylated MGMT glioblastoma: computational biological modeling predictions. J Neurooncol. 2021;153(3):393-402.
- Castro M, ..., Poornachandra G, et al. Cellworks Omics Biology Model (CBM) to predict therapy response and identify novel biomarkers for 5FU-based combination therapy in gastric cancer patients. J Clin Oncol. 2021;39(15_suppl):e16024.
- Marcucci G, ..., Poornachandra G, et al. Biosimulation using CBM identifies novel biomarkers to inform MEC-based combination therapy in refractory & relapsed AML patients. Blood. 2021;138(Suppl_1).
- Castro M, ..., Poornachandra G, et al. Biosimulation using CBM-based assessment of cytarabine and anthracycline response in AML. Blood. 2021;138(Suppl_1).
- Howard SC, ..., Poornachandra G, et al. Therapy biosimulation using CBM is predictive of individual AML patient probability of CR and OS: MyCare-023. Blood. 2021;138(Suppl_1).